

Not1MM User Manual

Mike Bridak K6GTE

June 2026

Contents

1	Not1MM	1
1.1	Not1MM: What is it?	1
1.2	The Why	1
1.3	Target Environment	1
1.4	Current State or Code Maturity	1
1.5	Known Issues	1
2	List of Should-Be-Working Contests	2
2.1	Other Not-So-Supported Contests	2
3	Installation	3
3.1	Not1MM Dependencies	3
3.2	Installation and Startup	3
3.2.1	Single line install	3
3.2.2	Multiple line install	3
3.2.3	Running the software	3
3.2.4	Python versions and rollbacks	4
4	Possible Post-Install Issues	4
4.1	Warning Message 1	4
4.2	Warning Message 2	4
5	Various File Locations	4
5.1	Data Files	4
5.2	Configuration Files	5
6	The Database	5
6.1	Why	5
6.2	The First One is Free	5
6.3	Why Limit Yourself?	5
6.4	Revisiting an Old Friend	5
7	Station Settings Dialog	6
7.1	Changing Station Information	6
7.2	Update Your CTY and SCP Files	6
8	Selecting a Contest	6
8.1	Selecting a New Contest	6
8.2	Selecting an Existing Contest as the Current Contest	7
8.3	Editing Existing Contest Parameters	7
9	Configuration Settings	8
9.1	Callsign Lookup	8
9.2	Soundcard	8
9.3	CAT Control	8
9.4	CW Keyer Interface	8
9.5	Cluster	9
9.6	N1MM Packets	10
9.7	Bands	10
9.8	Options	11
10	Logging Digital Contacts	13

11 Operating Multi-Multi	14
11.1 Network Settings for Multi-Multi	14
11.2 Contest Settings for Multi-Multi	15
12 Sending CW	16
12.1 Sending CW Macros	16
12.2 Auto CQ	16
12.2.1 Setting the delay	16
12.3 Sending CW Free Form	16
13 Editing Macro Keys	17
13.1 Macro Substitutions	17
13.2 Macro Use with Voice	17
13.2.1 Voice macro .wav files	17
13.2.2 Macro control of radio functions	18
14 Keyboard key bindings	19
14.0.1 And how to edit them	19
15 cty.dat and QRZ for Distance and Bearing	20
16 Other Uses for the Callsign Field	20
16.1 Frequency	20
16.2 Modes	20
16.3 Change Operator	20
17 The Windows	21
17.1 The Main Window	21
17.2 The Log Window	21
17.2.1 Editing a Contact	22
17.3 The Bandmap Window	24
17.4 The Check Partial Window	26
17.5 The Rate Window	27
17.6 The DXCC window	27
17.7 The Zone window	27
17.8 The Rotator Window (Work In Progress)	27
17.9 The Remote VFO Window	29
18 Cabrillo	30
19 ADIF	30
20 Recalculate Mults	30
21 ESM	31
21.1 Run States	32
21.1.1 CQ	32
21.1.2 Call Entered Send His Call and the Exchange	32
21.1.3 Empty Exchange Field Send AGN Till You Get It	32
21.1.4 Exchange Field Filled, Send TU QRZ and Logs it	33
21.2 S&P States	33
21.2.1 With His Call Entered, Send Your Call	33
21.2.2 If No Exchange Entered Send AGN	33
21.2.3 With Exchange Entered, Send Your Exchange and Log it	34
22 Call History Files	35

23	Creating your own Call History files	36
24	Contest Specific Notes	38
24.1	ARRL Sweekstakes	38
24.1.1	The Exchange Parser	38
24.1.2	The Exchange	38
24.2	ARRL 10M	38
24.2.1	If you're exchange is (RST + Serial No.)	38
24.2.2	If you're exchange is (RST + State/Province/ITU)	39
24.3	RAEM	39
24.4	RandomGram	39
24.5	UKEI DX	39
24.6	CWO Open Contest	39
24.7	DX-Pedition	39

1 Not1MM

The world's number 1 unfinished contest logger. *According to my daughter Corinna.

Not1MM's interface is a blatant rip-off of N1MM. It is NOT N1MM and any problem you have with this software should in no way reflect on their software.

1.1 Not1MM: What is it?

Not1MM attempts to be a usable ham/amateur radio contest logger. It's written in Python 3.10+ and uses Qt6 framework for the graphical interface and SQLite for the database.

1.2 The Why

Currently, this exists for my own personal enjoyment. I recently retired after 35+ years working for 'The Phone Company', GTE → Verizon → Frontier. And being a Gentleman of Leisure, I needed something to do in my free time. I am a casual tester and could not find any contesting software for Linux that I wanted to use. **Tucnak** is very robust and mature contest logger that is available at <http://tucnak.nagano.cz/>. It just wasn't for me.

1.3 Target Environment

The target operating system for this application is Linux. It may be able to run on other platforms such as BSD and Windows. But I don't have a way, or desire, to directly support them. I've recently purchased an M4 Mac Mini, so I'll probably put more effort into that platform as well.

1.4 Current State or Code Maturity

Not1MM is – at times – fairly stable. Recently, it would seem that I'm desperately trying to change that. The current focus of development is adding support for Multi-Multi contest operations. It is something that I have no practical experience in. So you can expect the same quality of code fit and finish.

1.5 Known Issues

- HamLib before 4.6.3 had a problem with sending CW and changing/reading the keying speed.
- wfview before version 2.2 has issues with frequency reporting and CW sending.

2 List of Should-Be-Working Contests

- General Logging
- DXPedition
- 10 10 Fall CW
- 10 10 Spring CW
- 10 10 Summer Phone
- 10 10 Winter Phone
- AGCW QRP
- ARI 40 80
- ARI DX
- ARRL 10M
- ARRL 160M
- ARRL DX CW, SSB
- ARRL Field Day
- ARRL RTTY Roundup
- ARRL Sweepstakes
- ARRL VHF
- CQ 160
- CQ WPX
- CQ World Wide
- CWOps CWT, CWO
- DARC Xmas
- DARC VHF
- Dutch PACC
- EA Majistad CW
- EA Majistad SSB
- EA RTTY
- ES FIELD DAY HF
- ES OPEN HF
- Estonian LL Cup
- EUDX
- Helvetia
- IARU Field day
- IARU HF
- ICWC MST
- Japan International DX
- K1USN Slow Speed Test
- Labre RS Digi
- LZ DX
- NAQP
- OK1WC
- Phone Weekly Test
- RAEM
- RAC Canada Day
- RandomGram
- REF CW, SSB
- RSGB 80M CC
- RSGB IOTA
- SAC CW, SSB
- SPDX
- SSA MT
- State QSO (limited)
- Stew Perry Top band
- TA VHF UHF
- UK/EI DX
- Weekly RTTY
- Work All Germany
- Winter Field Day

2.1 Other Not-So-Supported Contests

At present, there are no specific State QSO parties. There are, however, plug-ins for generic QSO parties that use either a signal report or serial number for the exchange.

3 Installation

3.1 Not1MM Dependencies

Install these software packages through your distribution package manager before continuing:

- Python 3.10+
- PyQt6
- libportaudio2
- libxcb-cursor0 (maybe... Depends on the distro)
- libxslt-devel (maybe... Depends on the distro)
- libxml2-devel (maybe... Depends on the distro)

3.2 Installation and Startup

3.2.1 Single line install

From the terminal type:

- `curl -LsSf uvx.sh/not1mm/install.sh | sh`

3.2.2 Multiple line install

Install uv as described at <https://docs.astral.sh/uv/>. Run the following line in the terminal:

- `curl -LsSf https://astral.sh/uv/install.sh | sh`

Instruct it to run Not1MM:

- `uvx not1mm@latest`

or install it with:

- `uv tool install not1mm@latest`

That's it... it will go out, fetch the latest version of Not1MM, setup a Python virtual environment, get all the needed python libraries, cache everything, and run Not1MM.

3.2.3 Running the software

Start Not1MM from the command line by typing:

- `not1mm`

The first time it runs will take a minute, but on subsequent launches it's lightning quick and will automatically check for updates. You will be prompted to download the latest version. A lovely non-AI generated icon may appear on the desktop that can be clicked to launch the application.

3.2.4 Python versions and rollbacks

If your Linux distro is not current it may be using an older version Python – perhaps 3.10. There can be significant performance benefits to using the latest Python release, but simply downloading and installing it will likely corrupt your system. The much safer approach is to run Not1MM inside a virtual environment with the desired Python version. To run Not1MM with Python 3.14, for example, you would type this line in the terminal:

- `uvx --python 3.14 not1mm@latest`

You can also rollback to an earlier version of Not1MM if the latest update has a bug (This happens a lot as we test in production). To run version 23.5.29 of the software with Python 3.10, for example, you would type:

- `uvx --python 3.10 not1mm==23.5.19`

Pow! Enjoy the pain...

4 Possible Post-Install Issues

4.1 Warning Message 1

WARNING: The script not1mm is installed in '/home/mbridak/.local/bin', which is not on PATH. Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

Try logging out and logging back in. Or reboot.

4.2 Warning Message 2

Warning: Ignoring XDG\SESSION_TYPE=wayland on Gnome.
Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
qt.qpa.plugin: Could not load the Qt platform plugin "xcb" even though it was found. This application failed to start because no Qt platform plugin could be initialized.
Reinstalling the application may fix this problem.

Use your package manager to load **libxcb-cursor0**.

If that is not an option, you can export an environment variable and launch the app like this:

```
mbridak@vm:~$ export QT_QPA_PLATFORM=wayland; not1mm
```

For a more permanent solution, you can place the line:

```
export QT_QPA_PLATFORM=wayland
```

in the .bashrc file of your home directory. Then after logging out and back in you should be able to launch it normally.

5 Various File Locations

5.1 Data Files

If your system has an XDG_DATA_HOME environment variable set, the database and CW macro files can be found there. Otherwise they will be in ~/.local/share/not1mm

5.2 Configuration Files

Configuration file(s) can be found at the location defined by

`XDG_CONFIG_HOME`

Otherwise they will be found in the home directory at

`~/.config/not1mm`

6 The Database

6.1 Why

The database holds... wait for it... data... I know, shocker right? A database can hold one or many contest logs. It also holds the station information, everything shown in the Station Settings dialog. You can have one database for the rest of your life. Filled with hundreds of contests you've logged. Or, you can create a new database to hold just one contest. You do you Boo.

6.2 The First One is Free

On initial launch, a database is created called 'ham.db'. This and all future databases are located in the data directory mentioned above.

6.3 Why Limit Yourself?

You can create a new database by selecting **File » New Database** from the main window, and give it a snazzy name. Why limit yourself? Hell, create one every day for all I care. You can manage your own digital disaster.

6.4 Revisiting an Old Friend

You can select a previously created database by selecting: **File » Open Database**.

7 Station Settings Dialog

After initial run of the program or when creating a new database, you will need to fill out the Station Settings dialog that will pop up.

The screenshot shows a 'Settings' dialog box with the following fields and values:

Call	K6GTE		
Name	Michael Bridak		
Address	2854 W Bridgeport Ave.		
Address			
City	Anaheim	State	CA
		Zip	92804
Country	United States		
Grid Square	Dm13at	CQ Zone	3
		ITU Zone	6
License	General	Latitude	33.8125
		Longitude	-117.9583
Station TX/RX		Power	
Antenna		Ant. Height	a.s.l.
ARRL Section	ORG		
Rover QTH			
Club	Americal Radio Relay League		
Email address	michael.bridak@gmail.com		

Buttons: Cancel, OK

You can fill it out if you want to. You can leave our friends behind 'Cause your friends don't fill, and if they don't fill. Well, they're no friends of mine.

You can fill. You can fill. Everyone look at your keys. *You had to be around in the 80's*

7.1 Changing Station Information

Station information can be changed any time by going to **File » Station Settings** and editing the information.

7.2 Update Your CTY and SCP Files

Before operating in a contest, you might want to update the CTY and SCP files. You can do this by choosing **FILE » Update CTY** and **FILE » Update MASTER.SCP**

8 Selecting a Contest

8.1 Selecting a New Contest

Select **File » New Contest**

New Contest

Select Contest Type for New Log

Log Type: CQ WPX CW

Start UTC: 2023-05-27 00:00:00

Operator: SINGLE-OP

Band: ALL

Power: LOW

Mode: CW

Overlay: N/A

Station: FIXED

Assisted: ASSISTED

Transmitter: ONE

Sent Exchange: # Exclude RST

Operators: K6GTE

Soapbox

Cancel OK

8.2 Selecting an Existing Contest as the Current Contest

Select **File » Open Contest**

Choose a Contest

Contest Name	Contest Start
cq_wpx_cw	2023-05-27 00:00:00
cq_wpx_ssb	2023-03-25 00:00:00

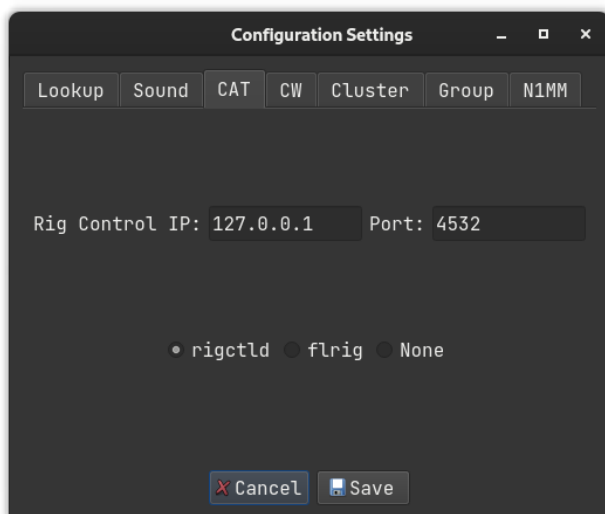
Cancel OK

8.3 Editing Existing Contest Parameters

Edit the parameters of a previously defined contest by selecting it as the current contest. Then select **File » Edit Current Contest**. Click 'OK' to save the new values and reload the contest. 'Cancel' to keep the existing parameters.

9 Configuration Settings

To setup CAT control, CW keyer, callsign lookups, and telnet cluster select **File » Configuration Settings**.



9.1 Callsign Lookup

Two services are supported: QRZ.com and HamQTH.com. Username and password are required so enter that information here. Note that a paid subscription to QRZ.com is required to use their lookup service.

9.2 Soundcard

Choose the sound output device for the voice keyer.

9.3 CAT Control

Under the CAT tab, you can choose one of the following:

- rigctld normally with an IP of 127.0.0.1 and a port of 4532
- flrig normally with an IP of 127.0.0.1 and a port of 12345
- None is always an option (but is it really?).

A small transceiver icon changes color to indicate the CAT status. Green: Connected; Red: Not connected; Grey: Neither.

9.4 CW Keyer Interface

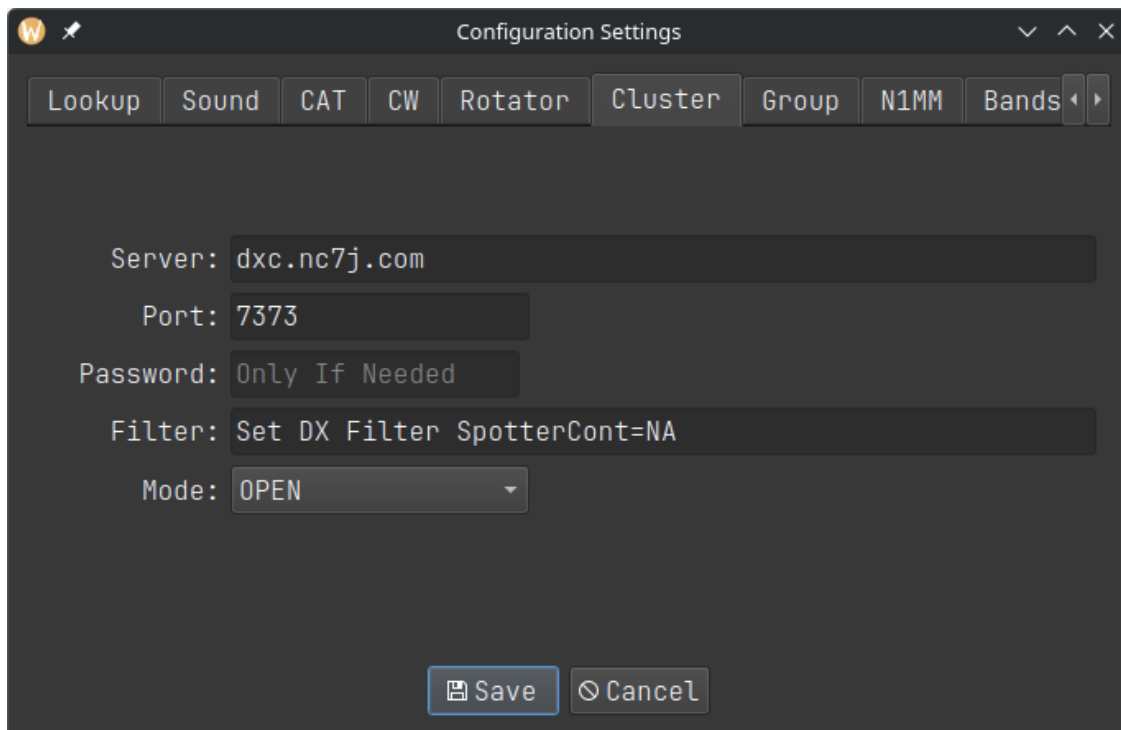
Under the CW tab, there are three options:

- cwdaemon normally uses IP 127.0.0.1 and port 6789
- pywinkeyer normally uses IP 127.0.0.1 and port 8000
- CAT can be used if your radio supports it. Morse characters are sent via rigctld.

Other CW configuration settings:

- Speed Increment for PgUp/PgDown: The PgUp and PgDown keys increment the keyer speed by the specified amount. The default step is ± 1 wpm with option to select steps anywhere in the range 1–10 wpm.
- Send 599 RST as: If a CW contest requires an RST report, the convention is to always use 599. This can be sent as either 5NN, 599, or ENN. 5NN is the most common.
- Serial Number Padding Character: Contest serial numbers can be optionally zero-padded. Choices for the padding character are 0, O, or T.
- Padding Length: Pad serial numbers with a specified number of padding characters. As an example, selecting the character T with a padding length of 3 sends the serial number 8 as TT8. Setting the padding length to 1 will result in no padding.
- Specify how to send characters 1, 9, 0 in contest serial numbers. When sent as cut numbers, the character A replaces 1, N replaces 9, and T replaces 0.

9.5 Cluster



Under the Cluster tab you can change the default AR cluster server, port, and filter settings used for the bandmap window. Setting the Mode to OPEN will display all spots on the selected band originating from all spotters anywhere in the world. Most users will want to enable a FILTER for spotters in the region in which they are located. The filter can be highly customized to select continents, countries, CQ zones, specific callsigns, and even Boolean filter combinations. To select spotting stations located in Europe, for example, the filter text can be set as:

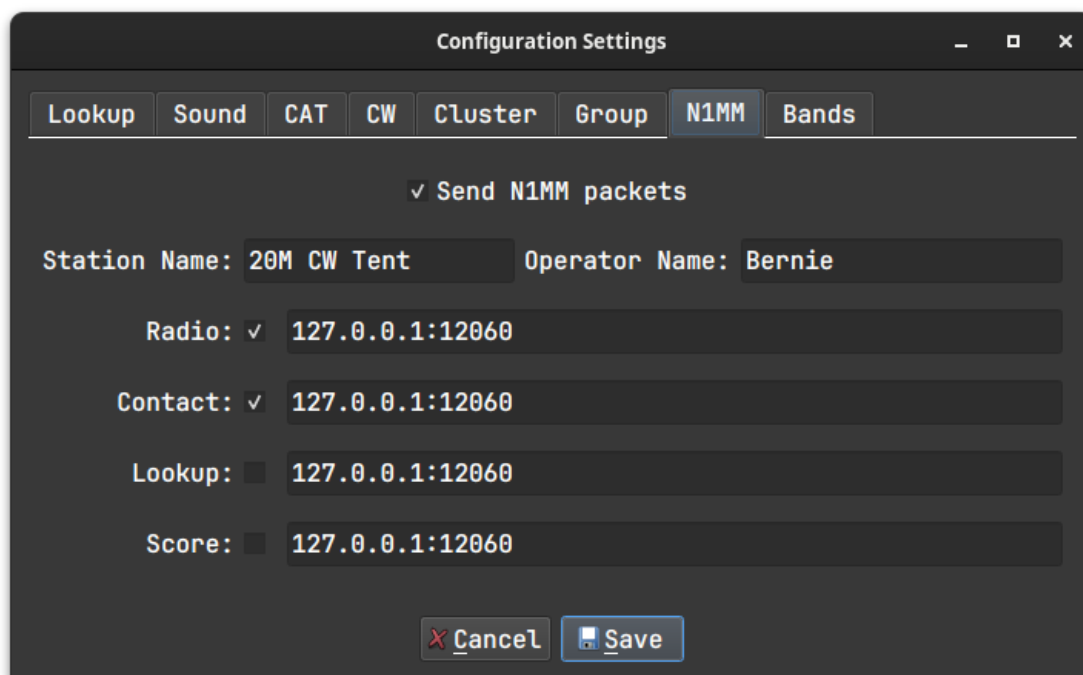
```
set/dx/filter spottercont=eu
```

Detailed information can be found at <http://www.nc7j.com/arcluster/arusermanual-6/AR-Cluster%20DX.html>.

Older versions of Python may not be able to handle the data rate of the telnet cluster stream, particularly during busy contests. This will cause spots on the bandmap to lose synchronization with the cluster. To remedy this, Not1MM can be run with the current version of Python inside a virtual environment as described in 3.2.4.

9.6 N1MM Packets

Work has started on N1MM UDP packets. So far just RadioInfo, contactinfo, contactreplace and contact-delete.



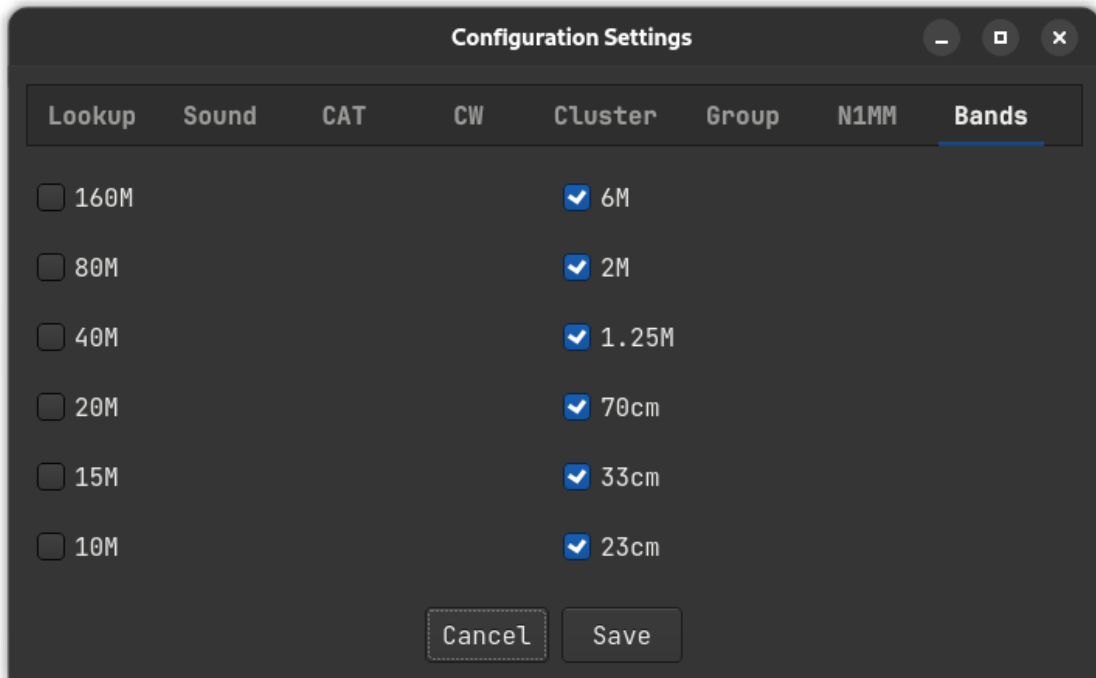
The screenshot shows a 'Configuration Settings' dialog box with the 'N1MM' tab selected. The dialog has a title bar with standard window controls. Below the title bar is a row of tabs: 'Lookup', 'Sound', 'CAT', 'CW', 'Cluster', 'Group', 'N1MM', and 'Bands'. The 'N1MM' tab is active. Inside the tab, there is a checkbox labeled 'Send N1MM packets' which is checked. Below this, there are four rows of input fields. The first row is 'Station Name: 20M CW Tent' and 'Operator Name: Bernie'. The second row is 'Radio: ✓ 127.0.0.1:12060'. The third row is 'Contact: ✓ 127.0.0.1:12060'. The fourth row is 'Lookup: 127.0.0.1:12060'. The fifth row is 'Score: 127.0.0.1:12060'. At the bottom of the dialog are two buttons: 'Cancel' and 'Save'.

Field	Value
Station Name	20M CW Tent
Operator Name	Bernie
Radio	✓ 127.0.0.1:12060
Contact	✓ 127.0.0.1:12060
Lookup	127.0.0.1:12060
Score	127.0.0.1:12060

When entering IP and corresponding Port, connect them with a colon ':' as shown above. You can enter multiple pairs on the same line by separating them with a space ' '.

9.7 Bands

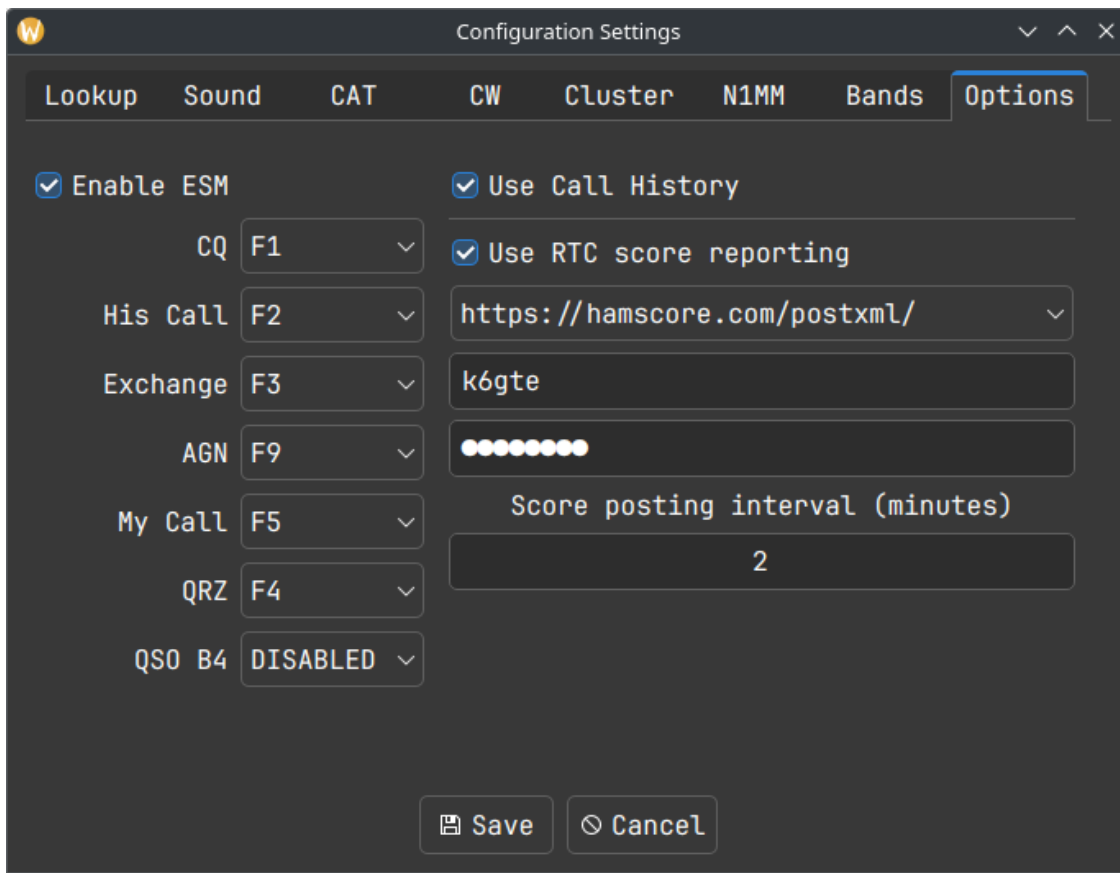
You can define which bands appear in the main window. Those with checkmarks will appear. Those without will not.



9.8 Options

A variety of features can be enabled in the Options tab:

- Use Enter Sends Message (ESM) and configure its function keys.
- Allow the program to access Call History info.
- Periodically send XML score to an online scoreboard.
- Clear the input fields in S&P mode when the VFO frequency changes.



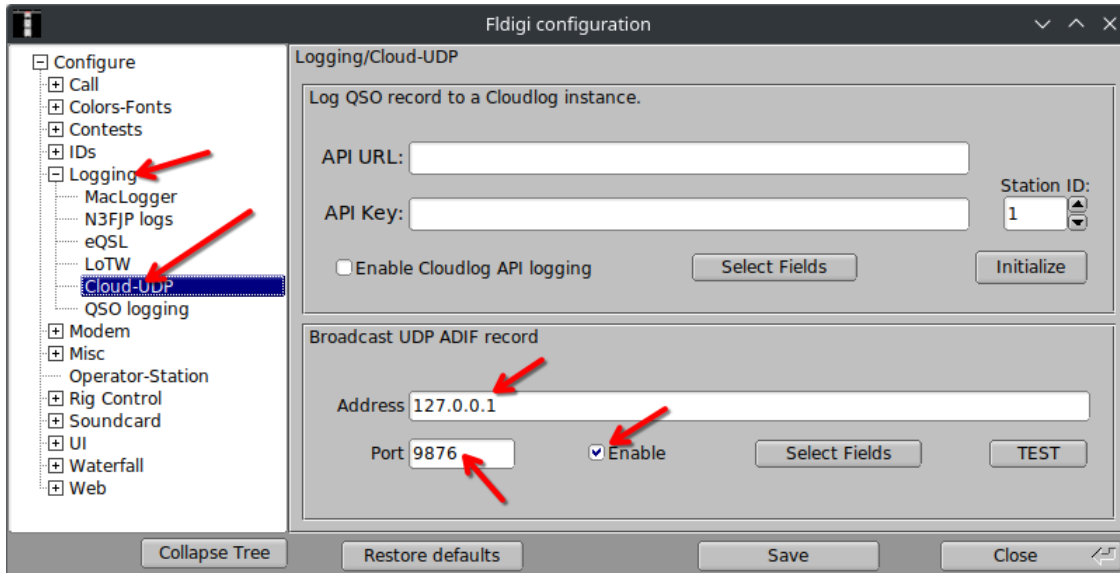
Three real-time scoreboards are supported:

- Real Time Contest Server at hamscore.com
- Contest Online ScoreBoard at contestonlinescore.com
- Live Contest Score at contest.run

10 Logging Digital Contacts

Not1MM listens for WSJT-X UDP traffic on the Multicast address 224.0.0.1:2237. This should work by default with Not1MM. That's good because I'm lazy.

Not1MM watches for fldigi QSOs by monitoring UDP traffic on 127.0.0.1:9876.

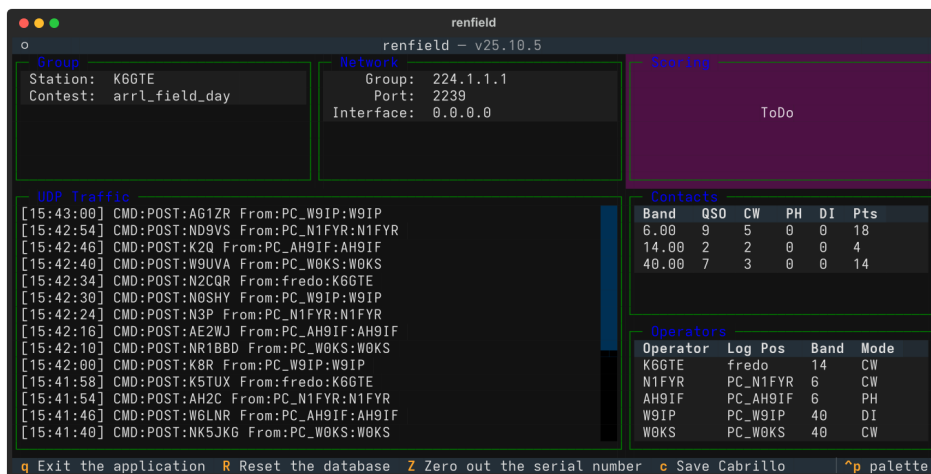


The F1–F12 function keys are sent to fldigi via XMLRPC. Fldigi is placed into TX mode, the message is sent, and a $\wedge$ r is appended to place it back into RX mode.

Unlike WSJT, fldigi needs configuration of some settings. The XMLRPC interface needs to be active. In fldigi's config dialog, go to **CONTESTS » General » CONTEST** and select Generic Contest. Make sure the Text Capture Order field says CALL EXCHANGE.

11 Operating Multi-Multi

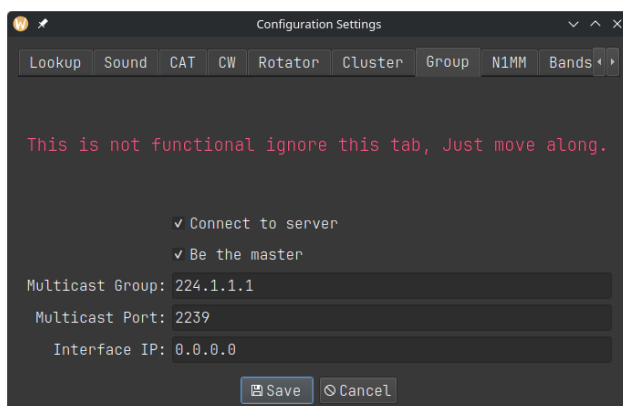
Work is underway on Multi-Multi contest operations. There is a companion project **renfield** <https://github.com/mbridak/renfield> that will be needed for this. The idea is to have a separate slave server running, preferably on another computer that's on the same network as the contesting PCs. This server will handle all the database CRUD operations. It will assign serial numbers, check for duplicate contacts, and generate a Cabrillo file.



This is a very lightweight terminal application that can be easily hosted on a Raspberry Pi or similar device. In a pinch it can be run alongside Not1MM on one of the contesting machines but use caution. You could even use this while operating alone as an automated backup. A copy of the log would be available in the event of a logging computer failure.

11.1 Network Settings for Multi-Multi

In the configuration dialog under the group tab, select Connect to server.



One computer needs to be the master station. The master station will tell the renfield server what contest is being run.

11.2 Contest Settings for Multi-Multi

In Contest Settings, if the Operator is set to "MULTI-OP" and the Transmitter category is not "ONE" or "SWL", Not1MM will ask the renfield server for serial numbers and dupe checks.

Log Type	HELVETIA
Start UTC	2025-09-28 00:00:00
Operator	MULTI-OP
Band	ALL
Power	LOW
Mode	CW
Overlay	N/A
Station	FIXED
Assisted	ASSISTED
Transmitter	UNLIMITED
Sent Exchange	#
	Exclude RST

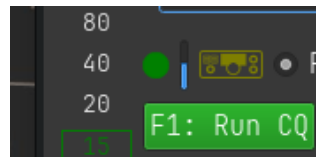
12 Sending CW

12.1 Sending CW Macros

Predefined CW messages can be sent with function keys F1–F12. See the following section on Editing Macro Keys.

12.2 Auto CQ

Pressing ‘SHIFT-F1’ activates the Auto CQ mode. The program is placed in the Run state and the F1 macro is sent then resent after the Auto CQ delay interval. A visual indicator appears to the upper left of the F1 macro key to alert that Auto CQ is active. A small progress bar indicates when F1 will fire next.



12.2.1 Setting the delay

The Auto CQ delay interval can be changed in the ‘Options’ tab of the Configuration settings. This value is the sum of the F1 macro duration *plus* the desired pause time in which stations can respond. If the program is in S&P mode when Auto CQ is enabled, it will automatically switch to RUN mode.

12.3 Sending CW Free Form

Freeform text is sent by pressing **CTRL-SHIFT-K**. An entry field appears at the bottom of the window for typing from the keyboard terminal. The text is sent by either pressing **CTRL-SHIFT-K** or using the Enter key to close the entry field.

13 Editing Macro Keys

To edit the macro keys, choose **File » Edit Macros**. This will open the system default text editor and display the current active macros. The current operating mode (CW, SSB or RTTY) and contest will automatically select the corresponding macro file. Each contest and mode has its own unique copy of the macros – these are not global settings. When finished editing, save the file and close the editor.

Alternatively, macro files can be directly edited outside the not1mm program. On Linux, individual contest directories are located in `./local/share/not1mm/`. Refer to the **Data Files** section above.

After editing and saving the macro file, any changes are enabled by toggling between the ‘Run’ and ‘S&P’ states.

13.1 Macro Substitutions

The macro instruction set is shown below.

Macro	Substitution
{MYCALL}	Send my station callsign.
{HISCALL}	Send the callsign of the other station.
{SNT}	Send 5NN (cw) or 599 (ssb)
{SENTNR}	Send the SentNR field.
{EXCH}	Send the Exchange field; contest must be defined.
{PREVNR}	Send the previous serial number.
{OTHER1}	Send the SentNR/Name field without altering it.
{OTHER2}	Send the Comment field without altering it.
{LOGIT}	Log the contact after macro pressed.
{TX}	Switch to transmit mode (PTT ON) - NOTE: the ‘Sends’ macros will toggle PTT ON/OFF
{RX}	Switch to receive mode (PTT OFF).
{TX/RX}	Toggle between transmit/receive (PTT) mode.
{MARK}	Highlight the current call in the band map.
{SPOT}	Spot the current call to the cluster.
{RUN}	Change to Run mode.
{SANDP}	Change to S&P mode.
{WIPE}	Clear all the input fields.
#	Sends serial number.
{VOICE1} - {VOICE10}	Uses ‘rigctld’ to send voice macros stored in the radio.
RI:	Send Rig specific codes. See Macro control of radio functions.
^M	in a RTTY macro will be replaced with a newline character.

13.2 Macro Use with Voice

VOICE1–VOICE10

If you use rigctld and your radio supports it, you can use the macros VOICE1, VOICE2, etc to send the recorded audio messages stored in the radio.

13.2.1 Voice macro .wav files

The macros used with voice can also linked to .wav files. Specify the filename, but exclude the .wav extension. The filename must be lowercase and enclosed in brackets. For example ‘[cq]’ will play ‘cq.wav’, ‘[again]’ will play ‘again.wav’. The .wav files are stored in the data directory. See **Various data file locations** above for the location of these data files. For me, the macro ‘[cq]’ will play

```
/home/mbridak/.local/share/not1mm/K6GTE/cq.wav
```

The default .wav files are not the ones you will want to use. They sound like an idiot. Use a program like Audacity or similar to record new .wav files in your own voice.

NATO phonetic .wav files are also provided for each letter and number. Their use can be illustrated with an example from the CQWW SSB Contest operating from Zone 3. The callsign K5TUX is entered and the following macro key is activated: 'HISCALL SNT SENTNR'. The program then sends this sequence of audio phonetics: KILO FIVE TANGO UNIFORM X-RAY FIVE NINE NINE THREE.

13.2.2 Macro control of radio functions

Macros can also be used to send CAT/CI-V control codes to the radio. To make use of this feature, start the command with the letters "RI" and a colon, followed by the desired command. If the command is ASCII text (for example, for Yaesu radios), just enter the text after "RI:". For binary codes, enter hexadecimal values separated by spaces.

For example, to enable the auto-notch filter on a Yaesu FT-710, the command would be "RI:BC01;". To do the same on an Icom 7300, the command would be something like "RI:FE FE 94 E0 16 41 01 FD". Refer to the operating manual or online sources for the correct command syntax. Note that CAT/CI-V command macros are only available when either FLRig or rigctl (HamLib) are being used for transceiver control.

14 Keyboard key bindings

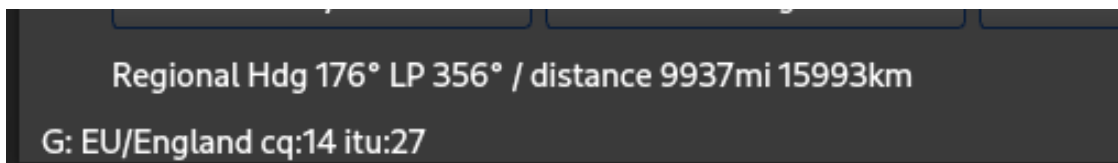
Key	Result
Esc	Stop CW sending and antenna rotation.
PgUp	Increase the CW sending speed.
PgDown	Decrease the CW sending speed.
Arrow-Up	Jump to the next spot above the current VFO cursor in the bandmap window (CAT Required).
Arrow-Down	Jump to the next spot below the current VFO cursor in the bandmap window (CAT Required).
TAB	Move cursor to the right one field.
Shift-Tab	Move cursor left One field.
SPACE	When in the callsign field, move the input to the next field in the exchange.
Enter	Submit entered data to the log, unless ESM is enabled.
F1-F12	Send (CW/RTTY/Voice) macros.
CTRL-S	Spot callsign to the cluster.
CTRL-M	Mark callsign in the bandmap window.
CTRL-G	Tune to a spot matching partial text in the callsign entry field (CAT Required).
CTRL-SHIFT-K	Open CW text input field.
CTRL=	Log the contact without sending the ESM macros.
CTRL-W	Clear the input fields of any text.
CTRL-R	Toggle the Run state.
CTRL-Q	Jump to last CQ frequency.
ALT-B	Toggle the bandmap window.
ALT-C	Toggle the Check Partial window.
ALT-L	Toggle the QSO Log window.
ALT-R	Toggle the Rate window.
ALT-S	Toggle the Statistics window.
ALT-V	Toggle the VFO window.
ALT-, (period)	Increase the VFO frequency by 20 Hz (cw); 100 Hz (ssb).
ALT-, (comma)	Decrease the VFO frequency by 20 Hz (cw); 100 Hz (ssb).

14.0.1 And how to edit them

Luckily now you don't have to be stuck with my preferred bindings. The key bindings can now be edited **File » Edit Keys**.

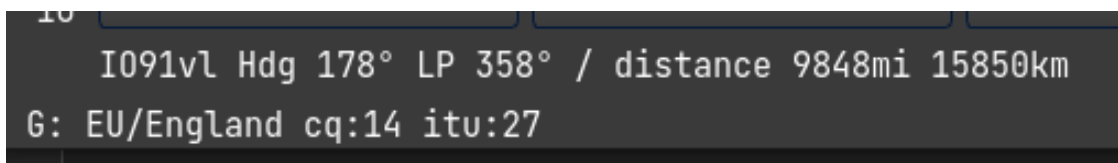
15 **cty.dat and QRZ for Distance and Bearing**

When a callsign is entered, a lookup is done in the cty.dat file to identify the country of origin, geographic center, CQ zone, and ITU region. Great circle calculations are then done to determine the heading and distance from the home grid square to the geographic center of the DX station. This information is displayed at the bottom left.



Regional Hdg 176° LP 356° / distance 9937mi 15993km
G: EU/England cq:14 itu:27

Next, a request is made to qrz.com for the grid square of the callsign. If there is a response, more accurate distance and bearing numbers are displayed. The 'Regional' text is replaced with the station's specific 6-digit grid square as shown below.



I091vI Hdg 178° LP 358° / distance 9848mi 15850km
G: EU/England cq:14 itu:27

16 **Other Uses for the Callsign Field**

The SPACE bar must be pressed after entering any of the following functions.

16.1 **Frequency**

Frequency Enter a frequency in kHz. This changes the active logging band. If CAT control is enabled, the specified frequency will change the transceiver VFO.

16.2 **Modes**

CW, SSB, RTTY Sets the mode being logged. If CAT control is enabled, the mode changes on the radio.

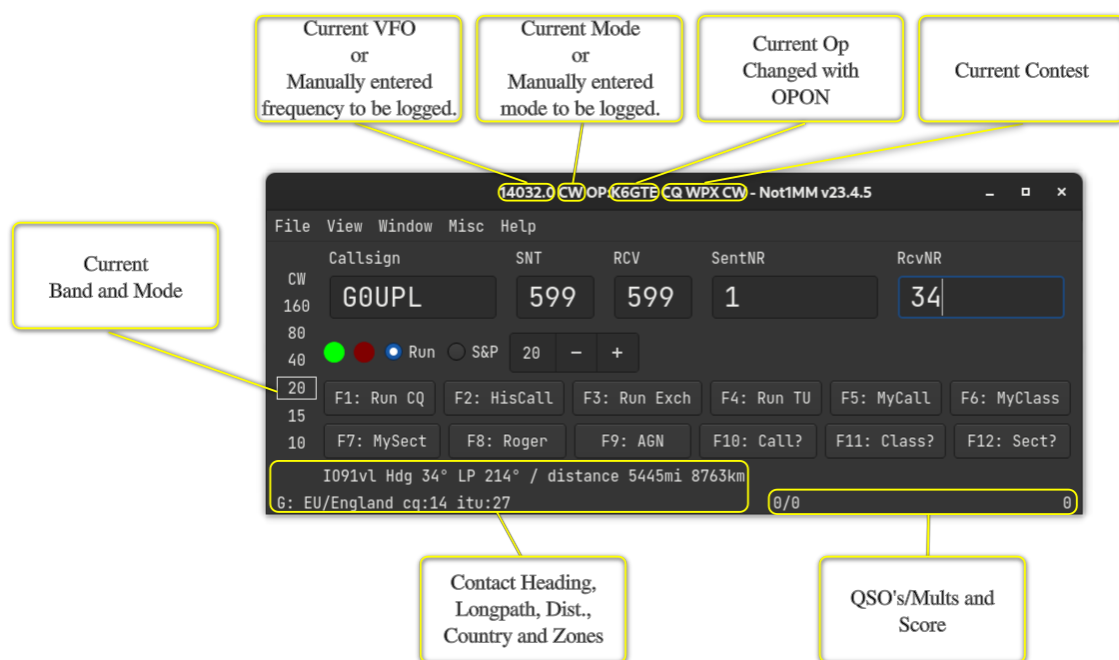
16.3 **Change Operator**

OPON Specify the operator currently logging.

Important: Press the SPACE bar after making any of the above entries.

17 The Windows

17.1 The Main Window



17.2 The Log Window

Window » Log Window

The Log Window gets updated automatically when a contact is entered. The top half is a chronological list of all contacts.

Log Display											x	
YYYY-MM-DD HH:MM:SS	Call	Freq	Snt	Rcv	M1	ZN	M2	PFX	PTS			
2023-03-17 13:39:03	K5TUX	14032.25	599	599	4		K5	0				
2023-03-17 14:14:23	W1AW	14032.25	599	599	5		W1	0				
2023-03-17 14:37:00	C0FEE	14032.25	599	599	0		C0	0				
2023-03-17 16:47:09	W1AW	14032.25	599	599	5		W1	0				
2023-03-17 16:56:55	W2AW	14032.25	599	599	5		W2	0				
YYYY-MM-DD HH:MM:SS	Call	Freq	Snt	Rcv	M1	ZN	M2	PFX	PTS			
2023-03-10 20:10:38	K6GTE	14230.2	59	59	3		K6	0				
2023-03-15 14:23:45	K6GTE	14062.0	59	59	3		K6	0				

The bottom half of the Log Window displays logged contacts sorted by complete or partial matches to characters in the call entry field. The columns displayed in the log window depend on the current active contest.

17.2.1 Editing a Contact

W	14032.25	599	5
Call	Freq	Snt	
W	14062.0		5

Double-click a cell in the log window to edit its contents. Right-clicking on a cell brings up the edit dialog.

Edit Contact

Call

G0UPL

TimeStamp

2023-03-23 18:50:23

Rx

14250.0

Tx Freq

14250.0

Mode

USB

Contest

CQ-WPX-SSB

RST Sent

59

RST RCV

59

Country

G

Station Call

K6GTE

Name

QTH

Comment

Nr

143

Nr Sent

2

Points

3

Power

0

Zone

14

Section

Mult

☒

Band

14

Check

0

Prec

Mult

☐

WPX

G0

Exchange

Run 1/2

Mult

☐

Radio

Grid

Op

K6GTE

Misc

Rover QTH

Delete

Cancel

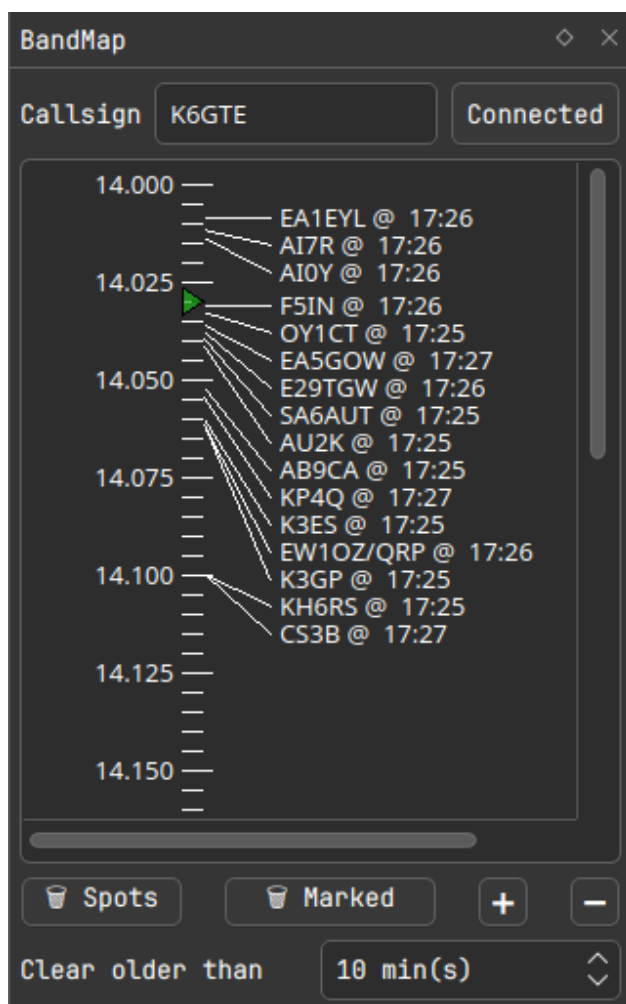
OK

You cannot directly edit the multiplier status of a contact. See the next section on recalculating multipliers. When editing a logged callsign, it's a good idea to check if other data such as the WPX field are still valid.

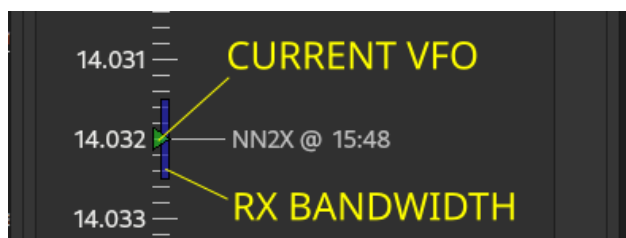
17.3 The Bandmap Window

Window » Bandmap

The bandmap displays worked stations and spots drawn from the Cluster node specified in the Configuration Settings. Enter your callsign and press the Connect button. When CAT is enabled, the bandmap will track with the VFO. This is still a work in progress.



The VFO frequency is indicated with a small green triangle above the tick marks. A vertical blue bar displays the receive filter bandwidth if available via CAT.



Clicking on a spot sets the VFO and populates the callsign field. Previously worked callsigns are displayed in gray. Callsigns marked with CTRL-M are color highlighted. A small icon after the callsign indicates the

mode as follows:

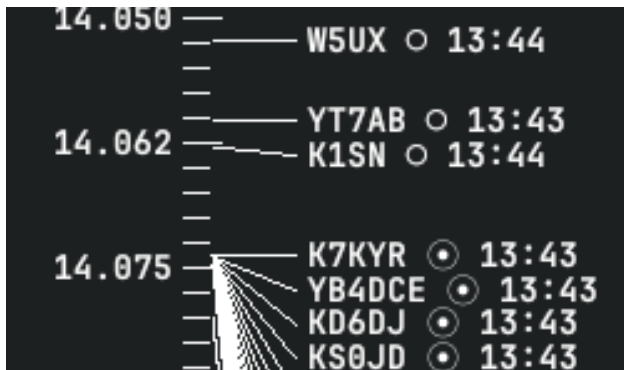
Primary Modes:

- ☐ CW
- ☒ FT*
- ☒ RTTY
- ☐ ^ Beacon
- ☐ @ Everything else

Secondary Modes:

- [P] POTA
- [S] SOTA

The icon is followed by the UTC time of the spot. The screenshot below shows CW and FT8 spots on the 20m band.



Right-clicking on a spot opens a menu to manage the entry:

Confirm: Refresh spot expiration timestamp and delete all other spots on this frequency (+- 0.1 kHz).

Mark: Mark/Unmark spot to work later and delete all other spots on this frequency.

Delete: Remove spot.

17.4 The Check Partial Window

Window » Check Partial Window

As a callsign is entered, the Check Window displays potential matches from the MASTER.SCP file, the local log, and recent telnet cluster spots. The MASTER.SCP column shows candidates with 3 or more matching characters indexed from the start of the callsign string. The local log and telnet columns match character strings of any length. Clicking on any of these candidate callsigns will automatically populate the callsign field.



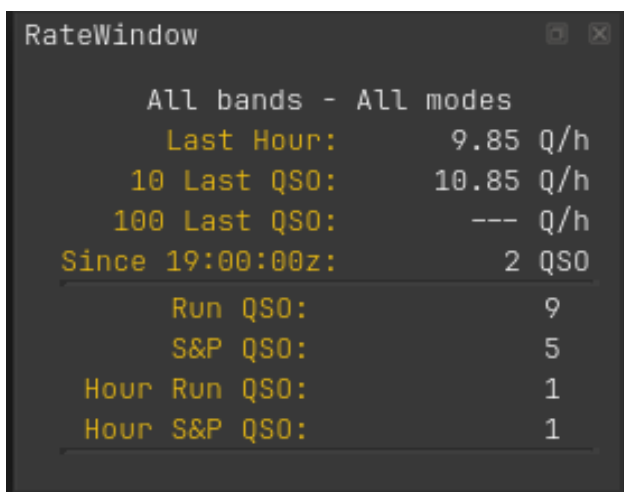
The screenshot shows a window titled "CheckWind" with a standard macOS-style title bar (minimize, maximize, close buttons). The window contains a table with three columns: "Log", "Master", and "Telnet". The "Log" column contains the text "LW2D0". The "Master" column contains a list of callsigns: "LW2DH", "LW2DHD", "LW2DJM", "LW2DLY", "LW2D0", "LW2D0D", "LW2DX", "LW2EDM", "LW2EIY", "LW2EKY", and "LW2HAB". The "Telnet" column contains the text "LW2D0".

Log	Master	Telnet
LW2D0	LW2DH	LW2D0
	LW2DHD	
	LW2DJM	
	LW2DLY	
	LW2D0	
	LW2D0D	
	LW2DX	
	LW2EDM	
	LW2EIY	
	LW2EKY	
	LW2HAB	

17.5 The Rate Window

Window » Rate Window

This window contains QSO rates and counts.



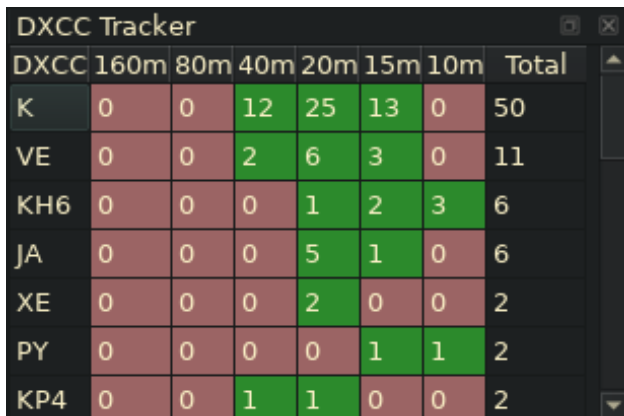
The RateWindow interface displays QSO rates and counts. It shows data for 'All bands - All modes' for the 'Last Hour', '10 Last QSO', '100 Last QSO', and 'Since 19:00:00z'. It also shows 'Run QSO', 'S&P QSO', 'Hour Run QSO', and 'Hour S&P QSO'.

All bands - All modes	
Last Hour:	9.85 Q/h
10 Last QSO:	10.85 Q/h
100 Last QSO:	--- Q/h
Since 19:00:00z:	2 QSO
Run QSO:	9
S&P QSO:	5
Hour Run QSO:	1
Hour S&P QSO:	1

17.6 The DXCC window

Window » DXCC

This window displays a table of worked DXCC entities arranged by band. When optional automatic scrolling is enabled, entering a callsign will scroll to the corresponding DXCC entity prefix.



The DXCC Tracker window displays a table of worked DXCC entities arranged by band. The table has columns for DXCC, 160m, 80m, 40m, 20m, 15m, 10m, and Total. The entities listed are K, VE, KH6, JA, XE, PY, and KP4.

DXCC	160m	80m	40m	20m	15m	10m	Total
K	0	0	12	25	13	0	50
VE	0	0	2	6	3	0	11
KH6	0	0	0	1	2	3	6
JA	0	0	0	5	1	0	6
XE	0	0	0	2	0	0	2
PY	0	0	0	0	1	1	2
KP4	0	0	1	1	0	0	2

17.7 The Zone window

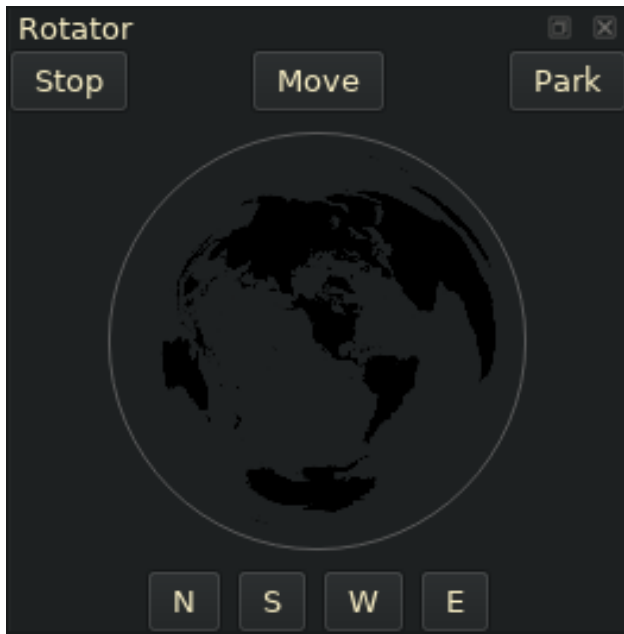
Window » Zone

This window functions the same way as the DXCC window, but for CQ or ITU zones. When automatic scrolling is enabled, check the appropriate radio button to select CQ or ITU zones as determined by the DX contest requirements.

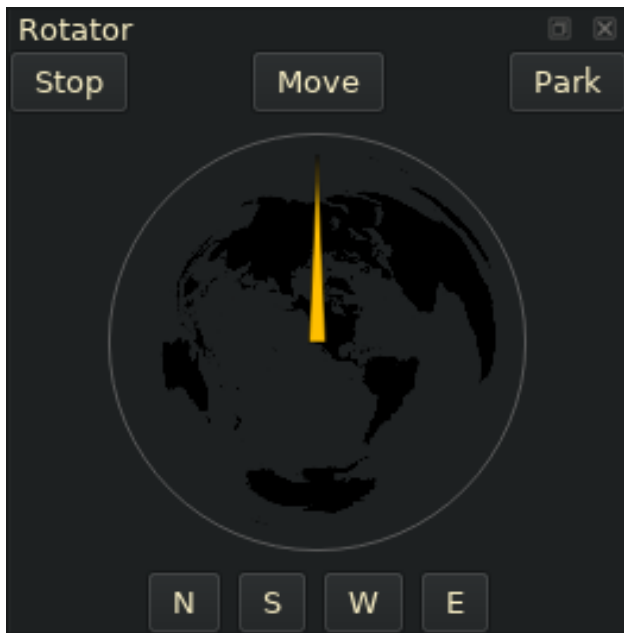
17.8 The Rotator Window (Work In Progress)

Window » Rotator

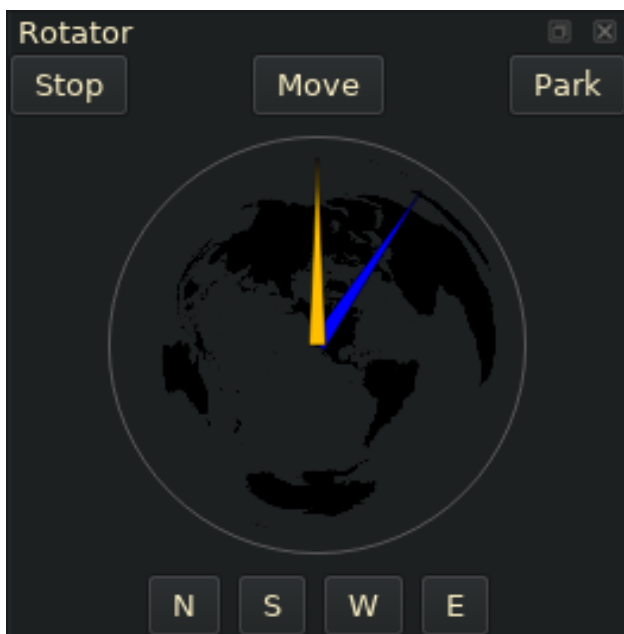
The Rotator window is a work in progress. The Rotator window relies on the functionality of the rotctld daemon. It connects to rotctld on address 127.0.0.1 and port 4533. You can change this if needed in the configuration dialog under the rotator tab. If started and there is no connection, you will see this:



Once there is a connection to rotctld, the current azimuth of the antenna is obtained and you will see a direction needle appear:



Once a call is entered and the bearing to contact is calculated you will see a needle appear:



At this time you can click on the 'Move' button to point your antenna at the contact. A list of other buttons follows below.

Move: Rotates the antenna at the target. Right-click to rotate to long-path (opposite) direction.

Stop: Stops the current movement.

Park: Parks the antenna.

N,S,W,E: Points the antenna to one of the 4 cardinal directions.

You can also move the rotator by clicking in the globe where you want the antenna to point.

17.9 The Remote VFO Window

When operating a remote station, the transceiver VFO can be controlled using inexpensive hardware in tandem with the Remote VFO window. A description of the hardware can be found at the following link:

https://github.com/mbridak/not1mm/blob/master/usb_vfo_knob/vfo.md

Communicate with it using the VFO Window: **Window » VFO**



18 Cabrillo

Click on **File » Generate Cabrillo**

The file will be placed in your home directory. The name will be in the format of:

`'StationCall'_'ContestName'_'CurrentDate'_'CurrentTime'.log`

So for me, it would look like:

`K6GTE\_CANADA-DAY\_2023-09-04\_07-47-05.log`

19 ADIF

File » Generate ADIF

Boom... ADIF

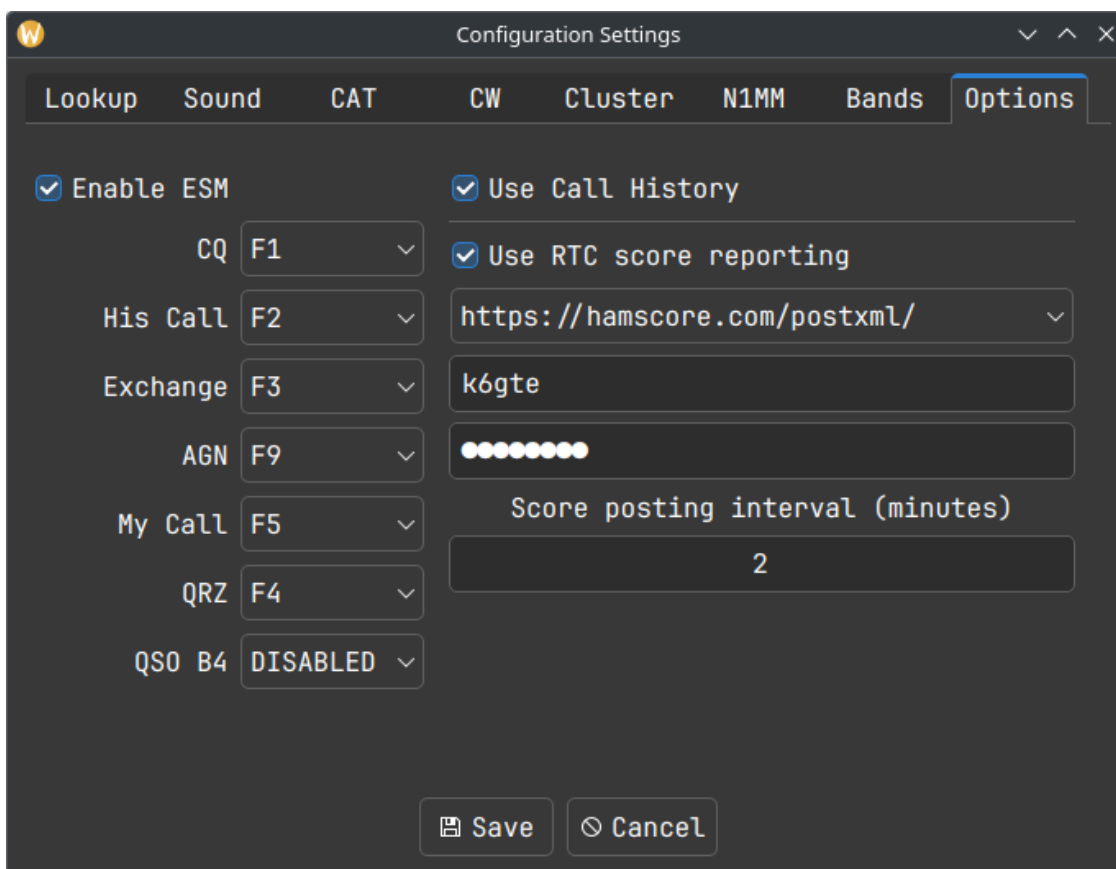
`'StationCall'_'ContestName'_'Date'_'Time'.adi`

20 Recalculate Mults

The program automatically updates the score as each new entry is logged. There may be a situation where a station has been logged incorrectly. This QSO can be easily edited as described above, but this may add or remove a multiplier. To remedy this and correct the score, a Recalculate Mults function is located in the Misc menu. The Cabrillo/ADIF files must be regenerated.

21 ESM

To turn on ESM or Enter Sends Message you can go to **FILE » Configuration Settings**



The screenshot shows the 'Configuration Settings' dialog box with the 'Options' tab selected. The 'Enable ESM' checkbox is checked. Below it, several function keys are mapped to specific actions: CQ (F1), His Call (F2), Exchange (F3), AGN (F9), My Call (F5), QRZ (F4), and QS0 B4 (DISABLED). To the right, the 'Use Call History' checkbox is also checked, and 'Use RTC score reporting' is checked with a URL of 'https://hamscore.com/postxml/'. Below this, the call sign 'k6gte' is entered, followed by a row of ten green dots representing function keys. The 'Score posting interval (minutes)' is set to 2. At the bottom, there are 'Save' and 'Cancel' buttons.

Function	Key
Enable ESM	☒
Use Call History	☒
Use RTC score reporting	☒
CQ	F1
His Call	F2
Exchange	F3
AGN	F9
My Call	F5
QRZ	F4
QS0 B4	DISABLED

URL: <https://hamscore.com/postxml/>

Call Sign: k6gte

Score posting interval (minutes): 2

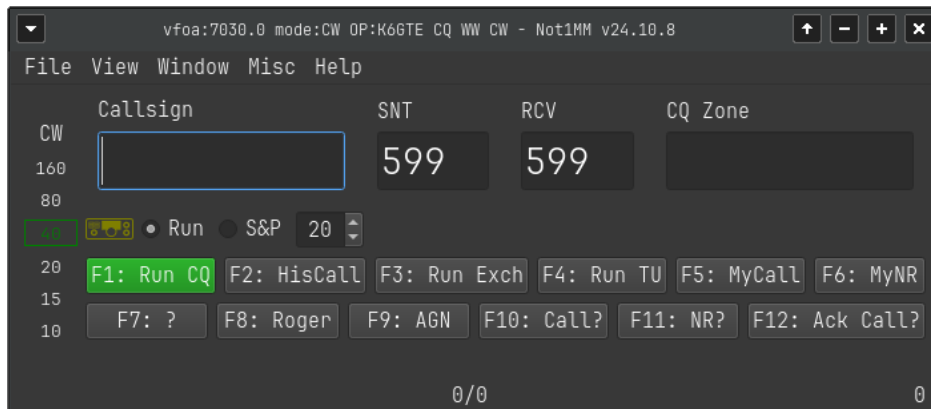
Buttons: Save, Cancel

Check the mark to Enable ESM and tell it which function keys do what. The keys will need to have the same function in both Run and S&P modes. The function keys will highlight green depending on the state of the input fields. The green keys will be sent if you press the Enter key. You should use the Space bar to move to another field.

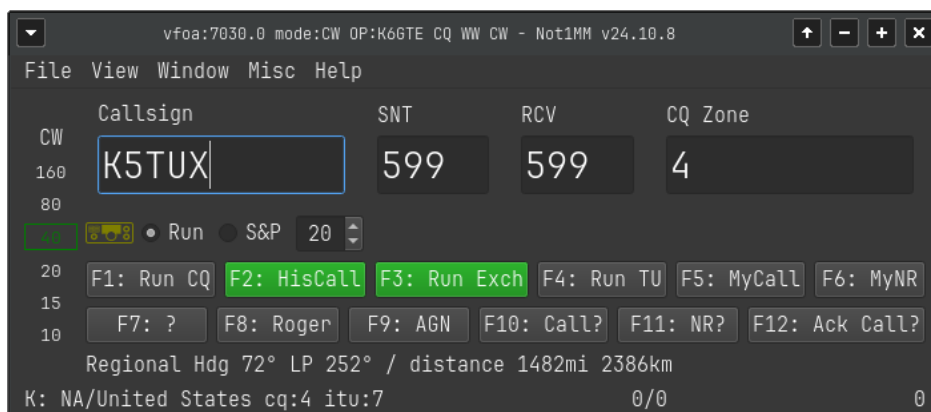
The contact will be automatically logged once all the needed info is collected and the QRZ (for Run) or Exchange (for S&P) is sent.

21.1 Run States

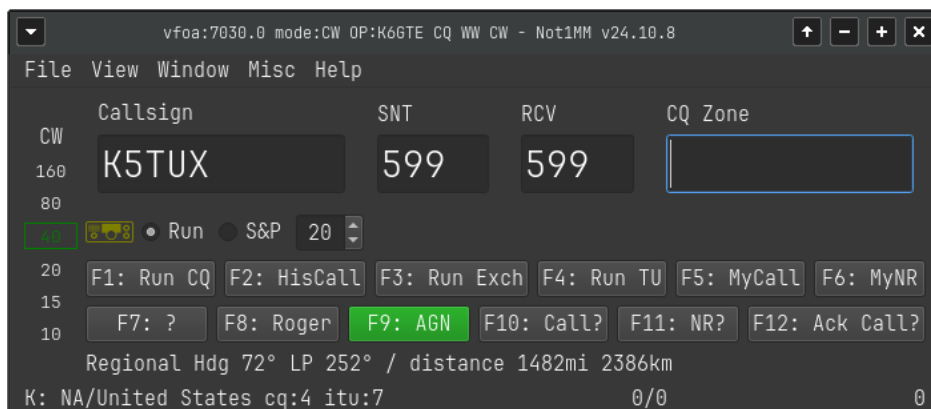
21.1.1 CQ



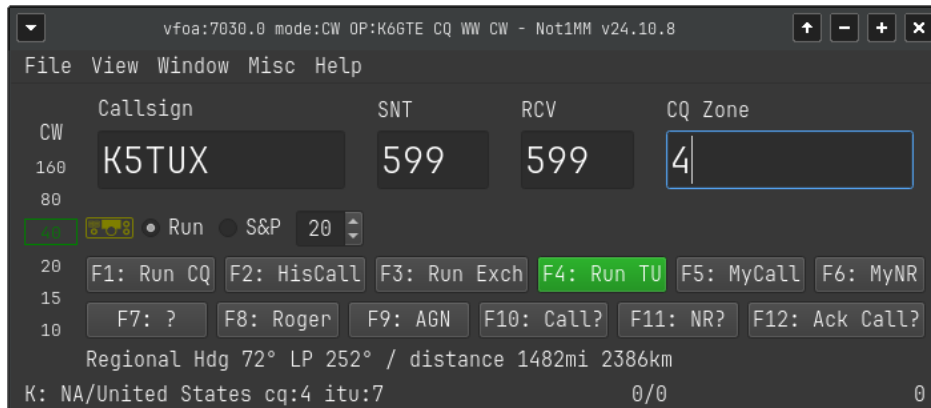
21.1.2 Call Entered Send His Call and the Exchange



21.1.3 Empty Exchange Field Send AGN Till You Get It

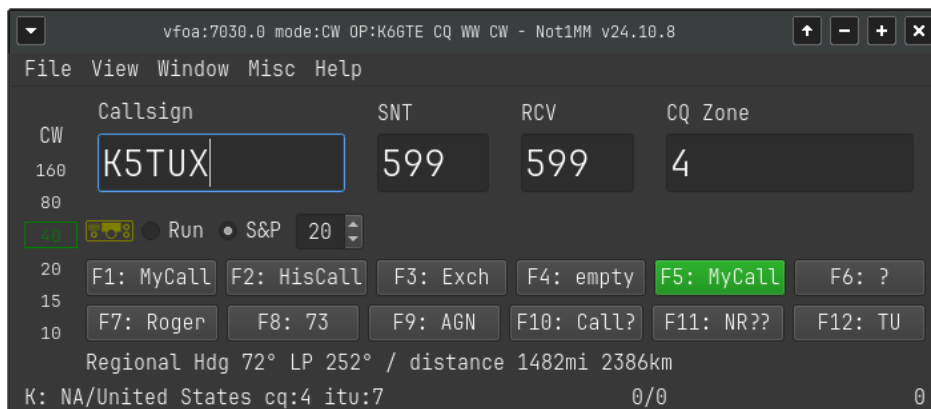


21.1.4 Exchange Field Filled, Send TU QRZ and Logs it

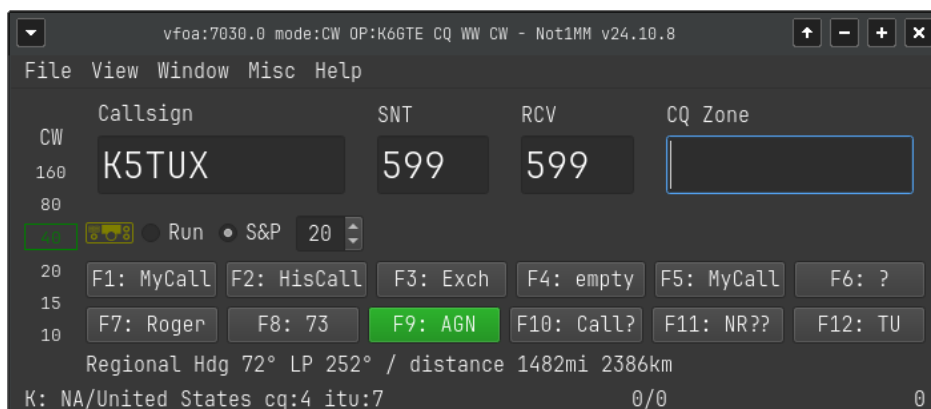


21.2 S&P States

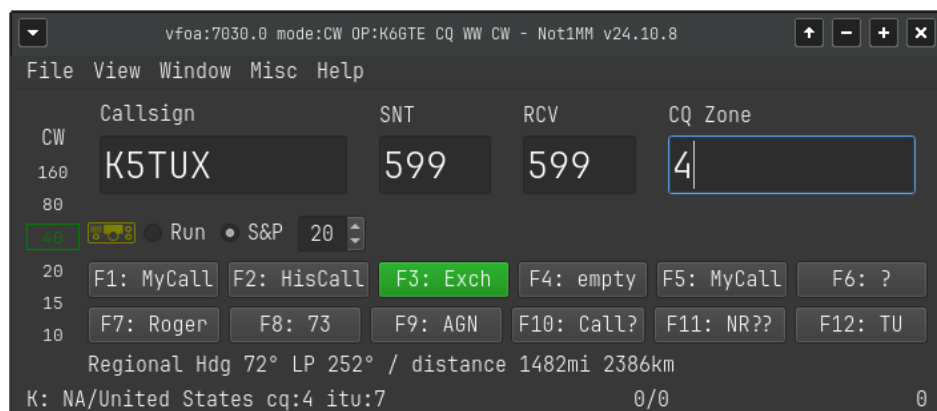
21.2.1 With His Call Entered, Send Your Call



21.2.2 If No Exchange Entered Send AGN

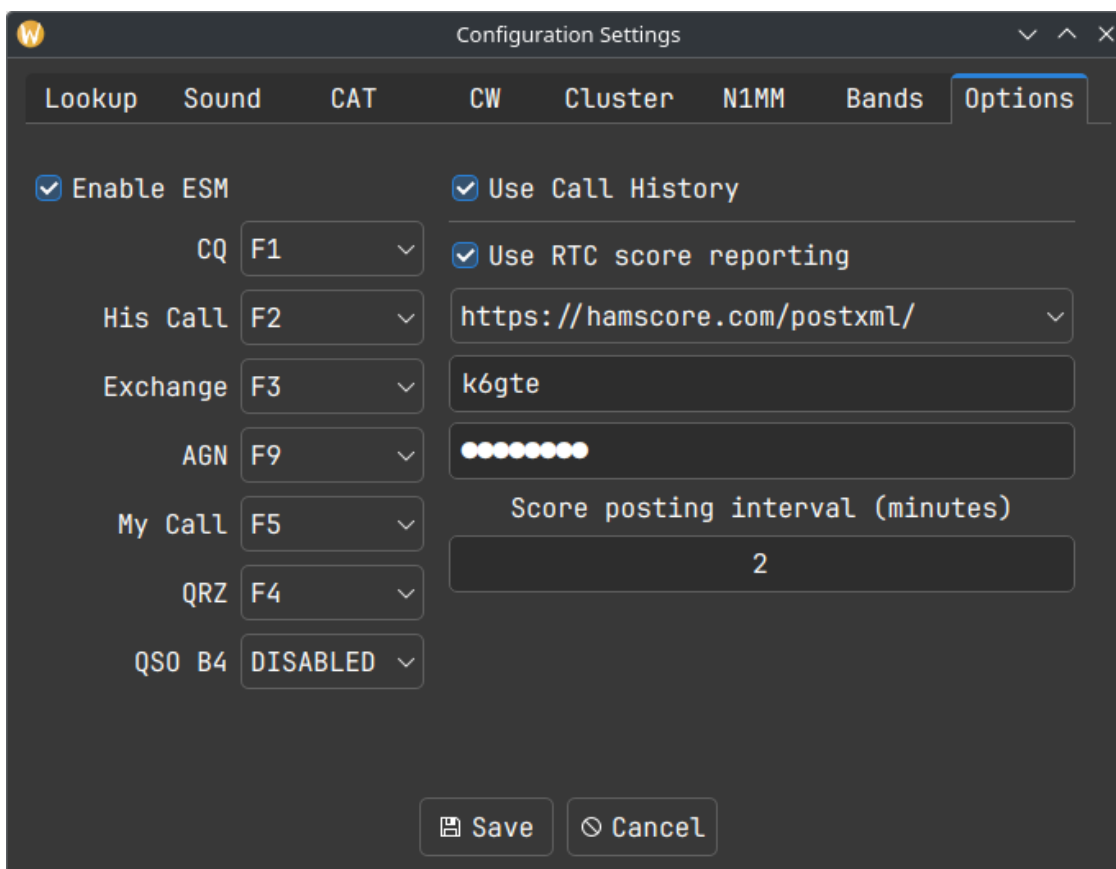


21.2.3 With Exchange Entered, Send Your Exchange and Log it



22 Call History Files

To use Call History files, go to **FILE » Configuration Settings**.



The screenshot shows the 'Configuration Settings' dialog box with the 'Options' tab selected. The 'Enable ESM' checkbox is checked. The 'Use Call History' checkbox is also checked. Below it, the 'Use RTC score reporting' checkbox is checked. The 'CQ' field is set to 'F1', 'His Call' to 'F2', 'Exchange' to 'F3', 'AGN' to 'F9', 'My Call' to 'F5', 'QRZ' to 'F4', and 'QS0 B4' to 'DISABLED'. The 'URL' field is set to 'https://hamscore.com/postxml/'. The 'k6gte' field is empty. The 'Score posting interval (minutes)' is set to '2'. The 'Save' and 'Cancel' buttons are at the bottom.

Field	Value
Enable ESM	☒
Use Call History	☒
Use RTC score reporting	☒
CQ	F1
His Call	F2
Exchange	F3
AGN	F9
My Call	F5
QRZ	F4
QS0 B4	DISABLED
URL	https://hamscore.com/postxml/
k6gte	
Score posting interval (minutes)	2

Place a check in the 'Use Call History' box. Call history files are very specific to the contest you are working. Example files can be obtained from <https://n1mmwp.hamdocs.com/mmfiles/categories/callhistory/> website. They have a searchbox so you can find the contest you are looking for. If you are feeling masocistic, you can craft your own. The general makeup of the file is a header defining the fields to be used, followed by lines of comma separated data.

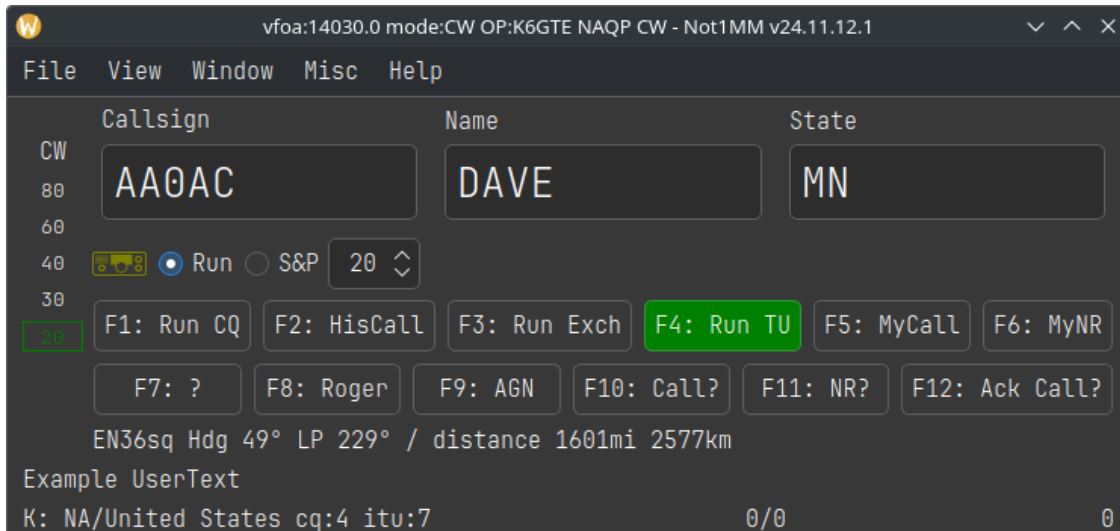
23 Creating your own Call History files

You can use **adif2callhistory** at <https://github.com/mbridak/adif2callhistory> to generate your own call history file from your ADIF files. An example file excerpt looks like:

```
!!Order!!,Call,Name,State,UserText,
#
# 0-This is helping file, LOG what is sent.
# 1-Last Edit,2024-08-18
# 2-Send any corrections direct to ve2fk@arrl.net
# 3-Updated from the log of Marsh/KA5M
# 4-Thanks Bjorn SM7IUN for his help.
# 5-Thanks
# NAQPCW
# NAQPRTTY
# NAQPSSB
# SPRINTCW
# SPRINTLADD
# SPRINTNS
# SPRINTRTTY
# SPRINTSSB
AAOAC,DAVE,MN,Example UserText
AAOAI,STEVE,IA,
AAOAO,TOM,MN,
AAOAW,DOUG,MN,
AAOBA,,TN,
AAOBR,,CO,
AAOBW,,MO,
```


The first line is the field definition header. The lines starting with a '#' are comments. Some of the comments are other contests that this file also works with. This is followed by the actual data. If the matched call has 'UserText' information, that user text is populated to the bottom left of the logging window.

So if one were to go to **FILE » LOAD CALL HISTORY FILE** and choose a downloaded call history file for NAQP and typed in the call AA0AC while operating in the NAQP, after pressing space, one would see:



Where the Name and State would auto-populate and the UserText info appears in the bottom left.

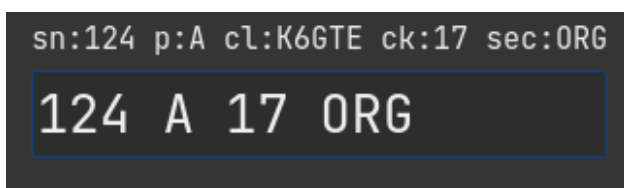
24 Contest Specific Notes

I found it might be beneficial to have a section devoted to weird quirky things about operating a specific contests.

24.1 ARRL Sweekstakes

24.1.1 The Exchange Parser

This was a pain in the tukus. There are so many elements to the exchange, and one input field aside from the callsign field. So I had to write sort of a 'parser'. The parser moves over your input string following some basic rules and is re-evaluated with each keypress and the parsed result will be displayed in the label over the field. The exchange looks like '124 A K6GTE 17 ORG', a Serial number, Precedence, Callsign, Year Licenced and Section. even though the callsign is given as part of the exchange, the callsign does not have to be entered and is pulled from the callsign field. If the exchange was entered as '124 A 17 ORG' you would see:



You can enter the serial number and precedence, or the year and section as pairs. For instance '124A 17ORG'. This would ensure the values get parsed correctly.

You do not have to go back to correct typing. You can just tack the correct items to the end of the field and the older values will get overwritten. So if you entered '124A 17ORG Q', the precedence will change from A to Q. If you need to change the serial number you must append the precedence to it, '125A'.

If the callsign was entered wrong in the callsign field, you can put the correct callsign some where in the exchange. As long as it shows up in the parsed label above correctly your good.

The best thing you can do is play around with it to see how it behaves.

24.1.2 The Exchange

In the 'Sent Exchange' field of the New Contest dialog put in the Precedence, Call, Check and Section. Example: 'A K6GTE 17 ORG'.

For the Run Exchange macro I'd put:

```
{HISCALL} {SENTNR} {EXCH}
```

24.2 ARRL 10M

The most confusing part of ARRL 10m is apparently how to setup your exchange.

24.2.1 If you're exchange is (RST + Serial No.)

You should edit your exchange macro keys to be:

```
{SNT} {SENTNR}
```

and a # character in the sent exchange field in the contest definition.



24.2.2 If you're exchange is (RST + State/Province/ITU)

You should edit your exchange macro keys to be:

`{SNT} {EXCH}`

and your state/prov/itu in the sent exchange field in the contest definition.

24.3 RAEM

In the New/Edit Contest dialog, in the exchange field put just your Lat and Lon. for me 33N117W. And in the exchange macro put '# {EXCH}'.

24.4 RandomGram

This plugin was submitted by @alduhoo. It reads a rg.txt file if it exists in the user's home directory to populate the next group in the sent exchange field.

24.5 UKEI DX

For the Run exchange macro I'd put '{SNT} # {EXCH}'

24.6 CWO Open Contest

Note: when completing the "Recd Number and Name" field, place a space between the received serial number and the name of the other operator. eg. "123 Fred". (Advance on spacebar is disabled for this field.)

24.7 DX-Pedition

This requested plugin is ment for operating a dxpedition. There's no scoring. No cabrillo generator. Be in Run mode with ESM. ESM mode opperates differently for this plug in than any other contest. The cursor does not advance to other fields when pressing enter.

Pressing enter with no call in the callsign field sends the CQ macro.

Pressing enter with a call in the callsign field sends HISCALL and the EXCHange macro.

Pressing enter again with an unaltered callsign in the field will send the QRZ macro and log the contact.

Pressing enter again with a different call than the one that was used two steps ago will assume you have corrected the call and send HISCALL macro and QRZ macro and log the contact.